

Life Cycle Assessment of PBAT-PLA Biodegradable Packaging Films

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Journal of Cleaner Production, 2023

Abstract

This study aims to quantify the environmental impacts of PBAT-PLA biodegradable packaging films using life cycle assessment. We applied Monte Carlo simulation to analyze uncertainty across end-of-life scenarios. Results show that industrial composting reduces global warming potential by 32% compared with landfill. However, the uncertainty remains high for marine degradation pathways.

Keywords: PBAT; PLA; Life Cycle Assessment; Monte Carlo; Composting

1. Introduction

Biodegradable plastics have attracted attention as alternatives.

2. Methods

We used a cradle-to-grave LCA framework and applied Monte Carlo with 10000 runs.

3. Results

Table 1. Environmental indicators by end-of-life scenario

Scenario	GWP (kg CO2-eq)	Energy (MJ)	Water (L)
Landfill	2.85	48.2	12.5
Incineration	2.10	61.7	9.8
Industrial composting	1.94	35.4	8.2
Anaerobic digestion	2.02	39.9	7.6

4. Discussion

The main limitation is the scarcity of primary data for marine degradation.

5. Conclusion

Results indicate that composting is the most favorable end-of-life option.